

LESSON 102 (1.5) GROUND

Lesson Objective

During this lesson, the applicant will learn how to obtain a weather briefing and complete a takeoff and landing data (TOLD) card. This lesson is intended to give the applicant a thorough understanding of the requirements for preflight planning.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Ask the student what they learned from the Pilot's Handbook of Aeronautical Knowledge chapter 10 - Weight and Balance.
 - Quiz the student on the FOM, including procedures for the following:
 - Time and scheduling
 - No-show appeals
 - Fatigue grounding
 - When to check ETA
 - Student questions.

References

- Advisory Circulars ([Search Tool](#))
- Aviation Weather Handbook. ([FAA-H-8083-28A](#))
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 10 - Weight and Balance.
 - Chapter 11 - Aircraft Performance.
 - Chapter 12 - Weather Theory.
 - Chapter 13 - Aviation Weather Services.
- Warrior III Pilot's Information Manual.
 - Section 05 - Performance.
 - Section 06 - Weight and Balance.

Instructor Guidance

- Keep this lesson engaging and fun, similar to the previous lesson. The student is still in the early stages of building their relationship with you and with aviation — enthusiasm here goes a long way.
- Encourage the student to explore supplemental weather applications in addition to Aviationweather.gov. Apps such as ForeFlight, Garmin Pilot, CloudAhoy & Windy are worth introducing early.
- Ensure the student is familiar with the weather briefing room, including the following:
 - Using the radio to obtain ATIS
 - Accessing updated W&B for aircraft and checking instructor availability
 - TOLD cards
 - The practice area overlay on the wall

- When working through the TOLD card for the first time, complete it together with the student rather than simply demonstrating. Walk through each field and explain its purpose — a student who understands why each item matters is far more likely to complete it accurately and independently going forward.
- Do not rush through weight and balance. This is a topic many students find intimidating at first. Use a real aircraft loading scenario and let the student work through the calculations with your guidance. If they make an error, ask them to find it themselves rather than correcting it immediately.

Lesson Content

Obtaining a Weather Briefing

METARs and TAFs

- Show the student how to decode METARs and TAFs and put the information on to the TOLD card.
 - Review issuance times and valid times.
 - Review what is included in the reports.
- Stress the importance of obtaining weather briefing prior to a flight (go/no-go decision).
- It is recommended to guide the student through a telephone/online briefing.
- Eventually a self-brief must progress beyond “weight and balance, METAR, TOLD card, and look outside”.
 - Types of weather briefings:
 - Outlook.
 - Standard.
 - Abbreviated.
 - In-flight.
 - For the purpose of this lesson, show the student how to obtain a weather briefing by calling the briefer or online.
 - Telephone:
 - Call 1-800-992-7433 → say “briefer” → say “Southern Florida”.
 - Give the briefer the flight plan information (use the reference on page 5 if needed) and request a standard weather briefing.
 - Have the weather products opened on aviationweather.gov during the briefing to aid student understanding.
 - Online:
 - Navigate to 1800wxbrief.com and have the student create an account.
 - Proceed to the plan & brief tab and input all of the required flight plan information (use the reference on page 4 if needed).
 - Click on route brief and select standard briefing.

Pilot Information Manual (PIM)

- Outline the different sections of the Pilot Information Manual.

- Discuss what can be found in each of the sections (GLENPWAAS).
 - **General.**
 - **Limitations.**
 - **Emergency procedures.**
 - **Normal procedures.**
 - **Performance.**
 - **Weight and balance.**
 - **Aircraft systems.**
 - **Aircraft servicing and handling.**
 - **Supplements.**

Weight and Balance

- TOLD Card:
 - Fill out all fields of a TOLD card with the student.
 - Explain the purpose of the TOLD cards:
 - Safety.
 - Preflight action ([91.103](#)).
- Weight and Balance (PIM Ch 06, PHAK Ch 10):
 - Introduce how to find arms and what they mean.
 - Show the student how to find CG and chart it on the envelope for takeoff and landing.
 - Effects of CG on performance:
 - Forward CG:
 - Decrease in cruise speed.
 - Increase in stall speed.
 - Increase in longitudinal stability.
 - Easier to recover from a stall.
 - Aft CG:
 - Increase in cruise speed.
 - Decrease in stall speed.
 - Decrease in longitudinal stability.
 - Harder to recover from a stall.
 - Effect of weight on performance:
 - Increase in weight:
 - Longer takeoff and landing distances.
 - Higher takeoff speeds.
 - Reduced rate and angle of climb.
 - Lower maximum altitude.
 - Shorter range.
 - Reduced cruising speed.
 - Higher stalling speed.

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- Higher approach and landing speed.
- Excessive weight on nose/tailwheel.
- Discuss ballast if necessary.
- Show the student how to find taxi fuel burn in the PIM.

Performance & Limitations

- Show the student how to find all the V-Speeds in the PIM.
- Discuss the atmospheric conditions' impact on performance.
 - Density and pressure altitude.
- Introduce the crosswind component chart to the student.
- Introduce the takeoff and landing charts (roll and 50-foot obstacle).

Post-Brief

- Questions from the student.
- Lesson grading.
- Preview the next lesson → 103.
- Assign homework:
 - Assign weight and balance practice problems.
 - Review METAR codes
 - Get familiar with Aviationweather.gov

Completion Standards

This lesson is complete when the applicant can:

1. Obtain a weather briefing by using 1-800-WX-BRIEF and complete a TOLD card with instructor guidance.
2. Understand the aircraft's performance charts with instructor guidance.
3. Calculate density altitude and apply it to performance calculations with instructor guidance.
4. Understand all environmental factors that affect performance.

[illegible]

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<u>Aircraft ID</u> Required	<u>Flight Rule</u> Required	<u>Flight Type</u> 	<u>No. of Aircraft</u> 1	<u>Aircraft Type</u> 	<u>Wake Turbulence</u> 	<u>Aircraft Equipment</u> 										
<u>Departure</u> Required	<u>Airport Info</u> Airport Info Area Brief	<u>Departure Date & Time</u> MM/DD/YYYY HHMM EST Required 1-120 Apply Minutes From Now		<u>Evaluate</u> Evaluate	<u>Cruising Speed</u> Required	<u>Level</u> Required	<u>Optimize</u> Optimize	<u>Surveillance Equipment</u> 								
<u>Route of Flight</u> DCT Map Plan				<u>Other Information (Optional)</u> 												
<u>Destination</u> Required	<u>Airport Info</u> Airport Info Area Brief	<u>Est Elapsed Time</u> HHMM Calculate		<u>Alternate 1 (Optional)</u> 	<u>Airport Info</u> Airport Info Area Brief	<u>Alternate 2 (Optional)</u> 	<u>Airport Info</u> Airport Info Area Brief									
<u>Fuel Endurance</u> HHMM	<u>Persons on Board</u> 	<u>Aircraft Color & Markings (Optional)</u> 		<u>Supplemental Remarks (Optional)</u> 		<u>Pilot In Command (Optional)</u> 										
<u>Emergency Radios</u> ☐ UHF ☐ VHF ☐ ELBA	<u>Survival Equipment</u> ☐ Polar ☐ Desert ☐ Maritime ☐ Jungle	<u>Jackets</u> ☐ Light ☐ Fluorescent ☐ UHF ☐ VHF	<u>Dinghies (Optional)</u> <table><tr><td><u>Number</u></td><td><u>Capacity</u></td><td><u>Color</u></td><td><u>Covered</u></td></tr><tr><td> </td><td> </td><td> </td><td>☐</td></tr></table>			<u>Number</u>	<u>Capacity</u>	<u>Color</u>	<u>Covered</u>				☐	<u>Pilot Contact Information</u> 		
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Route Brief File NavLog				Return Flight Plan Next Leg Clear												